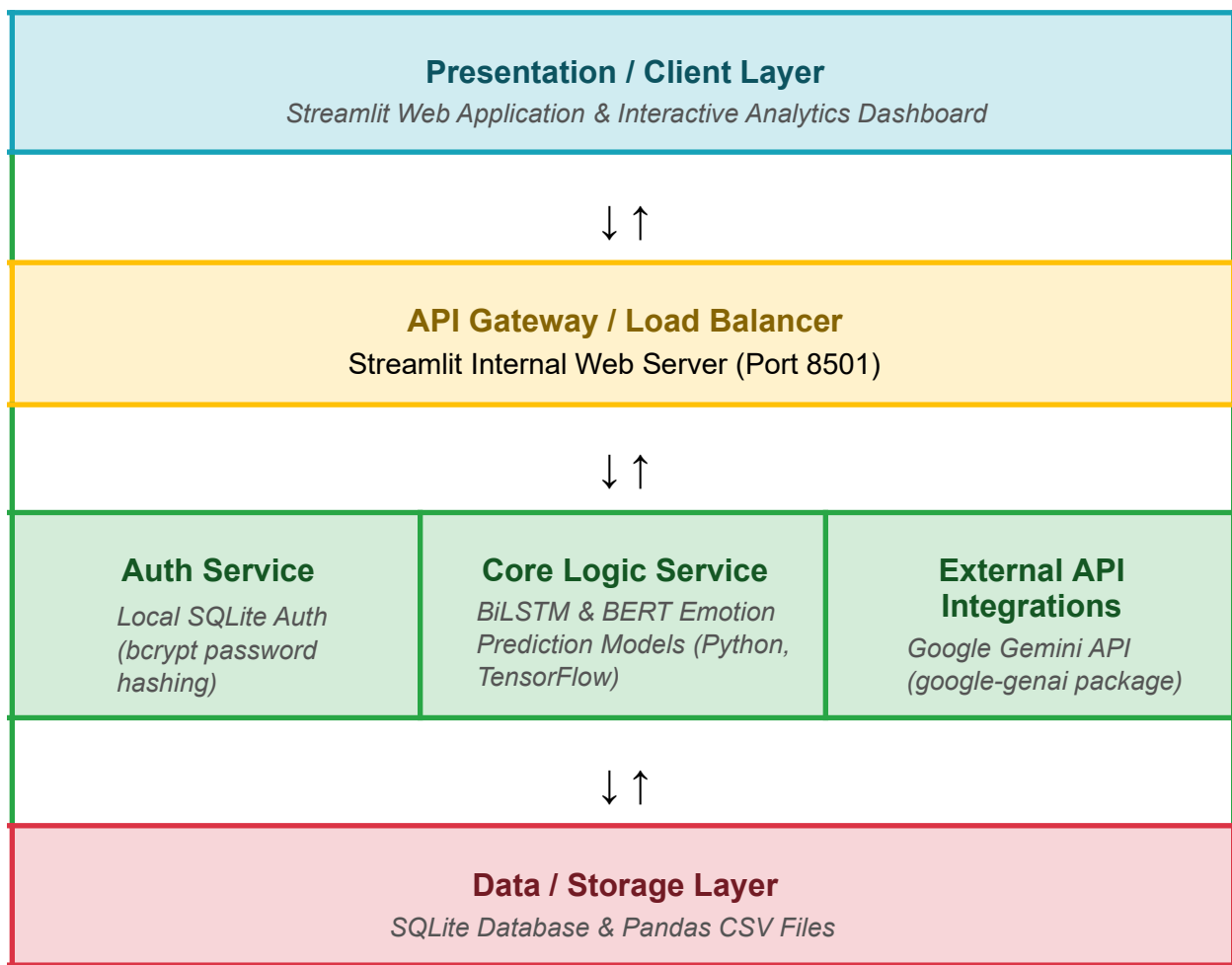


Solution Architecture

Date	6 July 2026
Team ID	xxxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the web-based user interface, captures student text input, and displays the AI responses and interactive Analytics Dashboard.	Streamlit, Plotly (for charts), HTML/CSS (via Streamlit markdown)
API Gateway / Web Server	Routes HTTP requests from the user's browser to the backend Python logic, and manages concurrent user session states.	Streamlit Internal Web Server (Tornado)
Core Logic & Microservices	Processes the student's text through deep learning models (BiLSTM/BERT) to predict their emotion, and passes this data to GenAI to formulate an empathetic response.	Python, TensorFlow/Keras, HuggingFace Transformers, Google Gemini API

Data / Storage Layer	Securely stores user login credentials and maintains a persistent history of all emotional states and AI interactions for analytics generation.	SQLite (Database), CSV files, Pandas
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